DSA to-do works

General Tasks

- find out the list of companies which are conducting the Drive placement riglets using DSA course
- find out the syllabus of gate for DSA.
- find out set of i/o for the following.
  1. Electricity bill
  2. Bus ticket
  3. Message to a friend
- Today when you started from home to college what are the various data structures you identified in your surroundings.

gate syllabus

1. Programming & Data Structures

- programming basics
- Recursion
- Arrays
- Stacks
- Queue
- linked lists
- Trees
- Binary Search Trees
- Binary Heaps
- Graphs

2. Algorithms

- searching - Linear search, Binary search
- sorting - Bubble sort, selection sort, Insertion sort, Merge sort, Quick sort, Heap sort
- Hashing
- Time and space complexity - Big-o(O), Omega (Ω), Theta (Θ)
- Algorithm Design Techniques - Divide and conquer, Greedy Algorithms, Dynamic programming
- Graph Algorithms - BFS, DFS, Minimum spanning tree, shortest path.

Companies List

Tier 1 (very high DSA)

Google, Microsoft, Amazon, Meta

Tier 2 (strong DSA)

Flipkart, Walmart, Intuit, Oracle, Apple, Paypal, NVIDIA, Visa, Uber, Atlassian, American Express, Adobe, Goldman Sachs, salesforce, Morgan Stanley, D.E. shaw

Indian product companies & Service Based companies

Indian product companies	Service Based companies (Basic to moderate DSA)
PhonePe	TCS
Razorpay	Infosys
Meesho	Wipro
Groww	Cognizant
CRED	Accenture
Freshworks	Capgemini
Zoho	HCL tech
Juspay	Tech Mahindra
Swiggy	IBM
Zomato	---

Input / Output Sets

1) Electricity bill

Input: Consumer Number, Customer Name, Previous Meter Reading, Current Meter Reading, Rate per unit, Fixed charge

Output: Units Consumed, Electricity Bill Amount, Due date, payment status

2) Bus ticket

Input	Output
passenger Name, Source, Destination, Date, Bus Type, Number of Tickets.	Ticket Number, Passenger Name, Seat Number, Fare, Bus Number, Journey Date

3) Sending a message to friend

Input	Output
Sender Name, Receiver Name, Message, Date, Time	Receiver, Message, Date, Time, status.

Surrounding Data Structures

Surrounding Object	Data Structure Identified
Mess token	object
plates	array
classroom benches	array
Books	array
Subjects	array
attendence register	array

Structure Size Analysis

Question: Create a structure variable student and estimate the size of the structure manually then write a C-program to find size of a structure using sizeof operator. Observe both the outputs. Are they same or different? Justify your answer.

C Program:

```
#include <stdio.h>

struct student
{
    int rollno;
    char grade;
    float marks;
};

int main()
{
    struct student s;
    printf("size of structure = %ld bytes\n", sizeof(s));
    return 0;
}
```

Manual Calculation:
int - 4
char - 1
float - 4
total = 9 bytes

Program Output:
size of structure = 12 bytes

Answer / Justification:
No, they are different. Because of structure padding (alignment). The compiler inserts extra unused bytes b/w members or at the end of the structure so that data is aligned in memory for faster access by the CPU.

Amazon Functions & Main Return Statement

Question: Open amazon website and find out 10 different function.

Answer:

1. Search products
2. Product Categories
3. Add to cart
4. Buy Now
5. Wishlist
6. Customer Reviews & Ratings
7. Order Tracking
8. Filter and sort
9. User login
10. Payment

Question: Return 1 or return 0 means what the main returns to whom.

Answer: It return the value to operating system (os).

- return 0: program executed successfully.
- return 1: program did not complete successfully. It contains general error/failure.

Time Complexity Calculations

Compute the time complexity using frequency count for the following:

1) Code Snippet:

```
for (i = 1; i <= 10; i++)
{
    printf("%d", i);
}
```

Frequency Count:
i = 1 : 1
i <= 10 : 11
i++ : 10
printf : 10
Total = 1 + 11 + 10 + 10 = 32

2) Code Snippet:

```
i = 1;
while (i <= 50)
{
    printf("hello");
    i++;
}
```

Frequency Count:
i = 1 : 1
i <= 50 : 51
printf("hello") : 50
i++ : 50
Total = 1 + 51 + 50 + 50 = 152

3) Code Snippet:

```
for (i = 1; i <= 20; i++)
{
    for (i = 1; i <= n; i++)
    {
        printf("hello");
        printf("world");
        printf("i = %d", i);
    }
}
```

Frequency Count:
Outer loop: 1 + 21 + 20 = 42
Inner loop: 1 + (n + 1) + 2n = 3n + 2
Total execution = 6n + 2

Java Hash Code

Question: In java every Object has hash code find out what is hash code and why it is being generated in java.

What is hash code?
It is an integer value that represents an object. It is mainly used to identify objects efficiently in hash-based collection such as HashMap, HashSet, HashTable.

Why it is generated?
To improve searching speed, helps compare objects efficiently, determine the bucket.

Example:

```
class Demo
{
    public static void main(String[] args)
```

{